

Dictionary Operations

Learning Objectives

- Perform advanced dictionary operations: iteration, merging, and comprehension
- Use dictionary methods to extract and transform data efficiently
- Solve real-world data processing problems using dictionaries

Engage (5-7 minutes)

A dictionary becomes truly powerful when you can iterate through it, merge multiple dictionaries, and transform data using comprehensions. These operations are essential for data analysis, configuration management, and building applications that process structured information. Today we unlock the full potential of dictionaries.

Explore (10-12 minutes)

Create two dictionaries: one with student names and marks in Math, another with the same students' marks in Science. Try to merge them so each student has both subjects. Then iterate through the merged dictionary to find students who scored above 80 in both subjects. Now use dictionary comprehension to create a new dictionary with only those high-performing students.

Explain (10-12 minutes)

Dictionary operations include: `keys()`, `values()`, and `items()` for iteration; `update()` or `|` for merging; `pop()` and `del` for removal; and `setdefault()` for safe insertion. Dictionary comprehensions follow the pattern `{key: value for item in iterable}`. You can merge dictionaries using `**dict1, **dict2` or the `|` operator. The `items()` method returns key-value pairs as tuples, enabling efficient iteration. Nested dictionaries allow hierarchical data representation.

Elaborate (8-10 minutes)

Build a comprehensive data processing system: create a dictionary of 10 students with nested dictionaries containing marks in 5 subjects. Write functions to: calculate each student's average, find the top performer per subject, identify the class topper, generate a grade report using dictionary comprehension, and merge two class dictionaries. Then implement a dictionary that caches previously computed results (memoization) for a recursive Fibonacci function. Discuss memory usage trade-offs.

Evaluate (5-8 minutes)

Given two dictionaries `d1 = {"a": 1, "b": 2}` and `d2 = {"b": 3, "c": 4}`, what is the result of merging them? Write a dictionary comprehension that creates a new dictionary from `range(10)` where keys are numbers and values are “even” or “odd”. Then write a function that inverts a dictionary (swaps keys and values), handling duplicate values appropriately.